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Program : BTech CSE (B16)

HDOC Day-13

Menu-Driven C Program Using Switch-Case

Problem Statement

Write a menu-driven C program using `switch-case` to input a **positive integer** and perform one of the following operations:

1. Check whether **all digits of the number are unique**.
 2. Find the **frequency of each digit from 0 to 9**.
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Algorithm

Step 1: Start

Step 2: Input

Read a positive integer `num`.

Step 3: Display Menu

1. Check whether all digits are unique
2. Find frequency of each digit from 0 to 9

Step 4: Input Choice

Read the user's choice into `choice`.

Step 5: Use Switch-Case

Case 1: Check Whether All Digits Are Unique

1. Initialize frequency array `freq[10] = {0}`.
2. Repeat while `num > 0`.
3. Extract the last digit:
`digit = num % 10`
4. Increment its frequency:
`freq[digit]++`
5. If `freq[digit] > 1`, set `unique = 0` and stop checking.
6. Remove the last digit:
`num = num / 10`
7. If `unique == 1`, display "Unique".
8. Otherwise, display "Not Unique".

Case 2: Find Frequency of Each Digit

1. Initialize `freq[10] = {0}`.
2. Repeat while `num > 0`.
3. Extract the last digit:
`digit = num % 10`
4. Increment:
`freq[digit]++`
5. Remove the last digit:
`num = num / 10`
6. Use a `for` loop from 0 to 9.
7. Display the frequency of every digit.

Default Case

Display "Invalid choice".

Step 6: Stop

Test Case Table

Test Case	Number Choice		Expected Output	Actual Output	Result
1	12345	1	Unique	Unique	Pass
2	12234	1	Not Unique	Not Unique	Pass
3	112333	2	freq_0=0, freq_1=2, freq_2=1, freq_3=3, others=0	Same as expected	Pass

Test Case	Number Choice	Expected Output	Actual Output	Result
4	10201 2	freq_0=2, freq_1=2, freq_2=1, others=0	Same as expected	Pass

C Program

```
#include <stdio.h>

int main()
{
    int num, choice, digit, i;
    int freq[10] = {0};
    int unique = 1;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    printf("\n1. Check whether all digits are unique");
    printf("\n2. Find frequency of each digit from 0 to 9");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            while(num > 0)
            {
                digit = num % 10;
                freq[digit]++;

                if(freq[digit] > 1)
                {
                    unique = 0;
                    break;
                }

                num = num / 10;
            }

            if(unique == 1)
                printf("Unique\n");
            else
                printf("Not Unique\n");

            break;

        case 2:
            while(num > 0)
            {
                digit = num % 10;
                freq[digit]++;
            }
    }
}
```

```

        num = num / 10;
    }

    for(i = 0; i <= 9; i++)
    {
        printf("freq_%d = %d\n", i, freq[i]);
    }

    break;

default:
    printf("Invalid choice\n");
}

return 0;
}

```

Compilation and Execution

Save the program as:

day13.c

Compile using the `cc` compiler:

`cc day13.c`

Execute:

`./a.out`

```

(kali@kali)-[~/Desktop]
$ cc day13.c

(kali@kali)-[~/Desktop]
$ ./a.out
Enter a positive integer: 12345

1. Check whether all digits are unique
2. Find frequency of each digit from 0 to 9
Enter your choice: 1
Unique

```

```

(kali@kali)-[~/Desktop]
$ ./a.out
Enter a positive integer: 12234

1. Check whether all digits are unique
2. Find frequency of each digit from 0 to 9
Enter your choice: 1
Not Unique

```

```
(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter a positive integer: 112333

1. Check whether all digits are unique
2. Find frequency of each digit from 0 to 9
Enter your choice: 2
freq_0 = 0
freq_1 = 2
freq_2 = 1
freq_3 = 3
freq_4 = 0
freq_5 = 0
freq_6 = 0
freq_7 = 0
freq_8 = 0
freq_9 = 0
```

```
(kali㉿kali)-[~/Desktop]
$ ./a.out
Enter a positive integer: 10201

1. Check whether all digits are unique
2. Find frequency of each digit from 0 to 9
Enter your choice: 2
freq_0 = 2
freq_1 = 2
freq_2 = 1
freq_3 = 0
freq_4 = 0
freq_5 = 0
freq_6 = 0
freq_7 = 0
freq_8 = 0
freq_9 = 0
```

Sample Execution — Operation 1

Input: 12345

Enter a positive integer: 12345

1. Check whether all digits are unique
2. Find frequency of each digit from 0 to 9
Enter your choice: 1

Unique

For 12234:

Enter a positive integer: 12234

1. Check whether all digits are unique
2. Find frequency of each digit from 0 to 9
Enter your choice: 1

Not Unique

Sample Execution — Operation 2

Input: 112333

Enter a positive integer: 112333

1. Check whether all digits are unique
2. Find frequency of each digit from 0 to 9
Enter your choice: 2

freq_0 = 0

```
freq_1 = 2
freq_2 = 1
freq_3 = 3
freq_4 = 0
freq_5 = 0
freq_6 = 0
freq_7 = 0
freq_8 = 0
freq_9 = 0
```

Evaluation Against Prepared Test Cases

Test Case 1

Number: 12345

Choice: 1

Digits → 1, 2, 3, 4, 5

No digit occurs more than once.

Expected Output: Unique

Actual Output: Unique

Result: PASS ✓

Test Case 2

Number: 12234

Choice: 1

Digit 2 occurs **twice**.

Expected Output: Not Unique

Actual Output: Not Unique

Result: PASS ✓

Test Case 3

Number: 112333

Choice: 2

- 1 occurs **2 times**
- 2 occurs **1 time**
- 3 occurs **3 times**
- All other digits occur **0 times**

Expected output matches actual output.

Result: PASS ✓

Test Case 4

Number: 10201

Choice: 2

- 0 occurs **2 times**
- 1 occurs **2 times**
- 2 occurs **1 time**
- Other digits occur **0 times**

Expected output matches actual output.

Result: PASS ✓

Result

The menu-driven C program was successfully written using **switch-case**. It correctly checks whether all digits are unique and calculates the frequency of digits from 0 to 9. **All prepared test cases passed successfully.**